A BILATERAL DOUGALL PROOF OF SUN'S CONJECTURE 5.2

Abstract. Let
$$g(x) = \frac{(74x+7)\Gamma(6x+1)}{(2x+1)4096^x\Gamma(3x+1)\Gamma(x+1)^3}.$$
We prove Zhi-Wei Sun's Conjecture 5.2 [3, Conjecture 5.2]:
$$\sum_{k\geq 0} g'(k) = 0, \qquad \sum_{k\geq 0} g''(k) = -48\pi^2, \qquad \sum_{k\geq 0} g'''(k) = 10752\zeta(3).$$
The proof deforms the Chu–Zhang acceleration of Dougall's nonterminating ${}_5F_4$ sum, bilateralizes the resulting nearly balanced series, and isolates its negative-index tail. The bilateral gamma quotient gives the constant, linear, and quadratic coefficients. The negative tail vanishes to order three and supplies the missing cubic correction; it reduces to the elementary Euler sum
$$\sum_{k\geq 0} \frac{H_k}{(2k+1)^2} = \frac{7\zeta(3) - \pi^2 \log 2}{4}.$$
This also explains structurally why the third derivative is the first order at which a new correction appears.

1. Statement and analytic setup

For an integer n, write $(z)_n = \Gamma(z+n)/\Gamma(z)$ whenever the quotient is defined.

Theorem 1.1. *For*
$$g(x) = \frac{(74x+7)\Gamma(6x+1)}{(2x+1)4096^x\Gamma(3x+1)\Gamma(x+1)^3}, \qquad x > -\frac{1}{6},$$
one has
$$\sum_{k=0}^{\infty} g(k) = 8,$$
$$\sum_{k=0}^{\infty} g'(k) = 0,$$
$$\sum_{k=0}^{\infty} g''(k) = -48\pi^2,$$
$$\sum_{k=0}^{\infty} g'''(k) = 10752\zeta(3).$$
In particular, all three identities in Sun's Conjecture 5.2 hold.

Set

$$Q(u) := \frac{\Gamma(6u+1)}{4096^u\Gamma(3u+1)\Gamma(u+1)^3}, \qquad S(u) := \sum_{k=0}^{\infty} g(k+u). \tag{1}$$

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The multiplication formula for the gamma function gives

$$\frac{Q(k+u)}{Q(u)} = \left(\frac{27}{64}\right)^k \frac{(u+\frac{1}{6})_k(u+\frac{1}{2})_k(u+\frac{5}{6})_k}{(u+1)_k^3}. \tag{2}$$

Consequently, for u in a fixed sufficiently small complex neighborhood of zero, the summands in $S(u)$ and all their derivatives of any prescribed finite order are bounded by

$$C_m\left(\frac{27}{64}\right)^k (1+k)^{C_m}(1+\log(2+k))^m.$$

Thus S is holomorphic near 0, differentiation is termwise, and it suffices to prove

$$S(u) = 8 - 24\pi^2u^2 + 1792\zeta(3)u^3 + O(u^4). \tag{3}$$

2. A specialized Chu–Zhang deformation

Following Chu and Zhang [1], define

$$\Omega(a;b,c,d,e) := \sum_{k=0}^{\infty}(a+2k)\frac{(b)_k(c)_k(d)_k(e)_k}{(1+a-b)_k(1+a-c)_k(1+a-d)_k(1+a-e)_k}. \tag{4}$$

Put

$$\begin{aligned}\mathcal{G}(a;b,c,d,e) := {}& \frac{\Gamma(1+a-b)\Gamma(1+a-c)\Gamma(1+a-d)\Gamma(1+a-e)}{\Gamma(b)\Gamma(c)\Gamma(d)\Gamma(e)}\\ &\times\frac{\Gamma(b+e-a)\Gamma(c+e-a)\Gamma(d+e-a)\Gamma(1+2a-b-c-d-e)}{\Gamma(1+a-b-c)\Gamma(1+a-b-d)\Gamma(1+a-c-d)}.\end{aligned} \tag{5}$$

Theorem 27 of Chu–Zhang [1, Theorem 27] states that, when $\Re(1+2a-b-c-d-e)>0$,

$$\begin{aligned}\Omega(a;b,c,d,e)-\mathcal{G}(a;b,c,d,e) = {}& \sum_{k=0}^{\infty}\frac{(e)_{3k}\prod_{x\in\{b,c,d\}}(x)_k}{\prod_{x\in\{b,c,d\}}(1+a-x)_{2k}}\\ &\times\frac{\prod_{\{x,y\}\subset\{b,c,d\}}(1+a-x-y)_k}{\prod_{x\in\{b,c,d\}}(x+e-a)_k}\nu_k(a;b,c,d,e).\end{aligned} \tag{6}$$

where

$$\begin{aligned}\nu_k(a;b,c,d,e) = {}& \frac{(a-c+2k)(a-e)}{a-c-e-k}\\ &-\frac{(c+k)(e+3k)(a-e)(1+a-b-d+k)}{(1+a-d+2k)(a-b-e-k)(a-c-e-k)}\\ &+\frac{(b+k)(c+k)(e+3k)(1+e+3k)(a-e)}{(1+a-b+2k)(1+a-c+2k)(1+a-d+2k)}\\ &\times\frac{(1+a-b-d+k)(1+a-c-d+k)}{(a-b-e-k)(a-c-e-k)(a-d-e-k)}.\end{aligned} \tag{7}$$

We specialize

$$a = \frac{1}{2}+3u, \qquad b=c=d=p:=\frac{1}{2}+u, \qquad e = a-\varepsilon. \tag{8}$$

The balancing parameter in (6) is then exactly ε.

Lemma 2.1. *For u near 0 and initially $\varepsilon > 0$, the specialization (8) satisfies*

$$\nu_k = -\frac{\varepsilon\, P(k+u,\varepsilon)}{8\big(2(k+u)+1-2\varepsilon\big)^3}, \tag{9}$$

where

$$P(t,\varepsilon) = 296t^3 + 324t^2 + 102t + 7 - \varepsilon(432t^2 + 264t + 24) + \varepsilon^2(168t + 20). \tag{10}$$

In particular,

$$\lim_{\varepsilon\to 0} \frac{\nu_k}{\varepsilon} = -\frac{74(k+u)+7}{8(2(k+u)+1)}. \tag{11}$$

Moreover, if

$$A_k(u,\varepsilon) := \frac{(a-\varepsilon)_{3k}(p)_k^6}{(1+2u)_{2k}^3(p-\varepsilon)_k^3}, \tag{12}$$

then

$$A_k(u,0) = \frac{Q(k+u)}{Q(u)}. \tag{13}$$

Proof. Substitution of (8) into (7), followed by a common denominator, gives (9). At $\varepsilon = 0$ one uses the exact factorization

$$296t^3 + 324t^2 + 102t + 7 = (2t+1)^2(74t+7),$$

which proves (11). The factor multiplying ν_k in (6) becomes precisely (12). Finally, triplication of $(a)_{3k}$ and duplication of $(1+2u)_{2k}$ yield

$$A_k(u,0) = \left(\frac{27}{64}\right)^k \frac{(u+\frac{1}{6})_k(u+\frac{1}{2})_k(u+\frac{5}{6})_k}{(u+1)_k^3},$$

which is (13) by (2). □

The ratio of consecutive terms in (12) tends uniformly to 27/64 for (u,ε) in a sufficiently small compact neighborhood of $(0,0)$. Therefore dominated convergence in (6) gives

$$E(u) := \lim_{\varepsilon\to 0^+} \frac{\Omega(a;p,p,p,a-\varepsilon) - \mathcal{G}(a;p,p,p,a-\varepsilon)}{\varepsilon} = -\frac{S(u)}{8Q(u)}. \tag{14}$$

3. Bilateralization and the negative tail

Introduce the bilateral series

$$\mathcal{B}(a;b,c,d,e) := \sum_{k\in\mathbb{Z}} (a+2k)\frac{(b)_k(c)_k(d)_k(e)_k}{(1+a-b)_k(1+a-c)_k(1+a-d)_k(1+a-e)_k}. \tag{15}$$

Dougall's bilateral ${}_5H_5$ summation, in the form obtained from van Diejen's formula [4, (1.2b)] by setting $z = a/2$ and $g_x = a/2 - x$, is

$$\mathcal{B}(a;b,c,d,e) = \frac{\Gamma(\Delta)\displaystyle\prod_{x\in\{b,c,d,e\}}\Gamma(1+a-x)\Gamma(1-x)}{\Gamma(a)\Gamma(1-a)\displaystyle\prod_{\{x,y\}\subset\{b,c,d,e\}}\Gamma(1+a-x-y)}, \qquad \Delta = 1+2a-b-c-d-e. \tag{16}$$

The convergence condition is $\Re\Delta > 0$.

Lemma 3.1. *For parameters where both sides are defined,*

$$\mathcal{G}(a;b,c,d,e) = R(a;b,c,d,e)\mathcal{B}(a;b,c,d,e), \tag{17}$$

where

$$R(a;b,c,d,e) = \frac{\sin(\pi b)\sin(\pi c)\sin(\pi d)\sin(\pi e)}{\sin(\pi a)\sin\pi(b+e-a)\sin\pi(c+e-a)\sin\pi(d+e-a)}. \tag{18}$$

On the path (8),

$$R(u,\varepsilon) = \frac{\cos^3(\pi u)\cos\pi(3u-\varepsilon)}{\cos(3\pi u)\cos^3\pi(u-\varepsilon)}. \tag{19}$$

Proof. Divide (5) by (16). All gamma factors cancel in complementary pairs, and $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ gives (18). Formula (19) follows immediately from (8). □

Let $\mathcal{N}$ denote the negative-index tail of (15). Replacing k by $-m-1$ and using $(x)_{-n} = (-1)^n/(1-x)_n$ gives

$$\mathcal{N}(a;b,c,d,e) = -\kappa\,\Omega(2-a;1+b-a,1+c-a,1+d-a,1+e-a), \tag{20}$$

where

$$\kappa = \frac{(b-a)(c-a)(d-a)(e-a)}{(1-b)(1-c)(1-d)(1-e)}. \tag{21}$$

For the path (8),

$$\kappa = \frac{8u^3\varepsilon}{(\frac{1}{2}-u)^3(\frac{1}{2}-3u+\varepsilon)}. \tag{22}$$

Chu–Zhang's reciprocal relation [1, Theorem 5] can be written as

$$\Omega(A;B,C,D,E) = \frac{A-E}{\Delta_*}\Omega(A_*;B_*,C_*,D_*,E), \tag{23}$$

where

$$\begin{aligned} \Delta_* &= 1+2A-B-C-D-E, \quad A_* = 1+2A-B-C-D, \\ B_* &= 1+A-B-C, \qquad C_* = 1+A-B-D, \qquad D_* = 1+A-C-D. \end{aligned} \tag{24}$$

Apply it to

$$A = \frac{3}{2}-3u, \qquad B=C=D=1-2u, \qquad E = 1-\varepsilon.$$

Since the balance equals ε, we obtain

$$\Omega\left(\frac{3}{2}-3u;1-2u,1-2u,1-2u,1-\varepsilon\right) = \frac{\frac{1}{2}-3u+\varepsilon}{\varepsilon}\,H(u,\varepsilon), \tag{25}$$

where

$$H(u,\varepsilon) := \Omega\left(1;\frac{1}{2}+u,\frac{1}{2}+u,\frac{1}{2}+u,1-\varepsilon\right). \tag{26}$$

Combining (20)–(25) gives the particularly simple identity

$$\mathcal{N} = -C(u)H(u,\varepsilon), \qquad C(u) := \frac{8u^3}{(\frac{1}{2}-u)^3}. \tag{27}$$

Since $\mathcal{B} = \Omega + \mathcal{N}$, Lemma 3.1 and (27) imply

$$\Omega - \mathcal{G} = (1-R)\mathcal{B} + C(u)H(u,\varepsilon). \tag{28}$$

This is the key decomposition: the entire negative tail carries the explicit factor $C(u) = 64u^3 + O(u^4)$.

4. The balanced limit

Set

$$D(u,\varepsilon) := \varepsilon\mathcal{B}(u,\varepsilon), \tag{29}$$

where $\mathcal{B}(u,\varepsilon)$ means the specialization (8) of (15). Formula (16) gives

$$\begin{aligned} D(u,\varepsilon) = {} & \frac{\Gamma(1+2u)^3\Gamma(\frac12-u)^3}{\Gamma(\frac12+3u)\Gamma(\frac12-3u)\Gamma(\frac12+u)^3} \\ & \times \frac{\Gamma(1+\varepsilon)^2\Gamma(\frac12-3u+\varepsilon)}{\Gamma(\frac12-u+\varepsilon)^3}. \end{aligned} \tag{30}$$

Thus D is analytic at $\varepsilon=0$, and

$$D_0(u) := D(u,0) = \frac{\Gamma(1+2u)^3}{\Gamma(\frac12+3u)\Gamma(\frac12+u)^3}, \tag{31}$$

while

$$\frac{\partial_\varepsilon D(u,0)}{D_0(u)} = d(u) := 2\psi(1) + \psi\left(\frac12 - 3u\right) - 3\psi\left(\frac12-u\right). \tag{32}$$

Here $\psi = \Gamma'/\Gamma$.

Define

$$\alpha(u) := \partial_\varepsilon R(u,0) = \pi\bigl(\tan(3\pi u) - 3\tan(\pi u)\bigr), \tag{33}$$

$$\beta(u) := \partial_\varepsilon^2 \log R(u,0) = -\pi^2\bigl(\sec^2(3\pi u) - 3\sec^2(\pi u)\bigr). \tag{34}$$

Since $R(u,0)=1$, we have

$$\partial_\varepsilon^2 R(u,0) = \alpha(u)^2 + \beta(u). \tag{35}$$

The value $H_0(u) := H(u,0)$ follows from Dougall's nonterminating ${}_5F_4$ summation (equivalently, from (6) when its last parameter equals the leading parameter):

$$H_0(u) = \frac{\Gamma(\frac32-u)^3\Gamma(\frac12-3u)}{\Gamma(1-2u)^3}. \tag{36}$$

Lemma 4.1 (Cancellation of the polar part)**.** *For u near zero,*

$$\alpha(u)D_0(u) = C(u)H_0(u). \tag{37}$$

Proof. Using $\Gamma(\frac32-u) = (\frac12-u)\Gamma(\frac12-u)$,

$$\frac{C(u)H_0(u)}{D_0(u)} = 8u^3\frac{\Gamma(\frac12-u)^3\Gamma(\frac12+u)^3\Gamma(\frac12-3u)\Gamma(\frac12+3u)}{\Gamma(1-2u)^3\Gamma(1+2u)^3}.$$

The reflection formula reduces the right-hand side to

$$\frac{8\pi\sin^3(\pi u)}{\cos(3\pi u)}.$$

Finally,

$$\tan(3x) - 3\tan x = \frac{8\sin^3 x}{\cos(3x)},$$

so the last expression equals $\alpha(u)$. ∎

Write

$$H_1(u) := \partial_\varepsilon H(u,0). \tag{38}$$

Expanding (28) at $\varepsilon = 0$, with $\mathcal{B} = D/\varepsilon$, gives

$$\frac{(1-R)\mathcal{B} + CH}{\varepsilon} = -\frac{\alpha D_0}{\varepsilon} - \alpha D_0 d - \frac{\alpha^2+\beta}{2} D_0 + \frac{CH_0}{\varepsilon} + CH_1 + O(\varepsilon).$$

Lemma 4.1 cancels the two polar terms. Hence

$$E(u) = -D_0(u)\left(\alpha(u)d(u) + \frac{\alpha(u)^2+\beta(u)}{2}\right) + C(u)H_1(u). \tag{39}$$

A double use of the gamma duplication formula gives

$$Q(u)D_0(u) = \frac{1}{\pi^2}. \tag{40}$$

Combining (14), (39), and (40) yields the exact master identity

$$\boxed{S(u) = \frac{8}{\pi^2}\left(\alpha(u)d(u) + \frac{\alpha(u)^2+\beta(u)}{2}\right) - 8Q(u)C(u)H_1(u).} \tag{41}$$

It holds initially for $0 < u < \delta$; all terms on the right have a finite limit as $u \to 0^+$, which is all that is needed for the Taylor expansion of the holomorphic function S.

5. The cubic tail and an Euler sum

From (26),

$$H(u,\varepsilon) = \sum_{k=0}^{\infty}(1+2k)\frac{(\frac{1}{2}+u)_k^3(1-\varepsilon)_k}{(\frac{3}{2}-u)_k^3(1+\varepsilon)_k}. \tag{42}$$

The series, its first ε-derivative, and the first u-derivative of that ε-derivative converge uniformly for u in a small real neighborhood of zero. Differentiating at $\varepsilon = 0$ gives

$$H_1(u) = -2\sum_{k=0}^{\infty}(1+2k)\left(\frac{(\frac{1}{2}+u)_k}{(\frac{3}{2}-u)_k}\right)^3 H_k, \tag{43}$$

where $H_0 = 0$ and $H_k = \sum_{j=1}^{k} 1/j$. At $u = 0$,

$$H_1(0) = -2\sum_{k=0}^{\infty}\frac{H_k}{(2k+1)^2}. \tag{44}$$

Lemma 5.1. *One has*

$$\sum_{k=0}^{\infty}\frac{H_k}{(2k+1)^2} = \frac{7\zeta(3) - \pi^2\log 2}{4}. \tag{45}$$

Consequently,

$$H_1(0) = \frac{\pi^2\log 2 - 7\zeta(3)}{2}. \tag{46}$$

Proof. The generating function $\sum_{k\geq 0} H_k z^k = -\log(1-z)/(1-z)$ gives

$$\begin{aligned}\sum_{k=0}^{\infty}\frac{H_k}{(2k+1)^2} &= \int_0^1 \frac{\log x\,\log(1-x^2)}{1-x^2}\,\mathrm{d}x \\ &= \frac{1}{4}\int_0^1 t^{-1/2}\frac{\log t\,\log(1-t)}{1-t}\,\mathrm{d}t.\end{aligned} \tag{47}$$

Let $\mathrm{B}(a,b)=\Gamma(a)\Gamma(b)/\Gamma(a+b)$. The last integral is

$$\frac{1}{4}\lim_{b\to 0^+}\partial_a\partial_b \mathrm{B}(a,b)\big|_{a=1/2}.$$

As $b\to 0$,

$$\partial_a \mathrm{B}(a,b)=-\psi_1(a)+b\left((\gamma+\psi(a))\psi_1(a)-\frac{1}{2}\psi_2(a)\right)+O(b^2).$$

Therefore

$$\lim_{b\to 0^+}\partial_a\partial_b \mathrm{B}(a,b)=(\gamma+\psi(a))\psi_1(a)-\frac{1}{2}\psi_2(a).$$

Using

$$\psi\left(\frac{1}{2}\right)=-\gamma-2\log 2,\qquad \psi_1\left(\frac{1}{2}\right)=\frac{\pi^2}{2},\qquad \psi_2\left(\frac{1}{2}\right)=-14\zeta(3),$$

we obtain (45), and (46) follows from (44). □

6. Completion of the proof

The elementary expansions at $u=0$ are

$$\alpha(u)=8\pi^4u^3+O(u^5),\tag{48}$$

$$\beta(u)=2\pi^2-6\pi^4u^2+O(u^4),\tag{49}$$

$$d(u)=4\log 2+O(u^2),\tag{50}$$

$$C(u)=64u^3+O(u^4),\qquad Q(u)=1+O(u),\qquad H_1(u)=H_1(0)+O(u).\tag{51}$$

The bilateral gamma part of (41) is therefore

$$\frac{8}{\pi^2}\left(\alpha d+\frac{\alpha^2+\beta}{2}\right)=8-24\pi^2u^2+256\pi^2\log 2\,u^3+O(u^4).\tag{52}$$

By (46), the negative-tail term is

$$\begin{aligned}-8Q(u)C(u)H_1(u)&=-512H_1(0)u^3+O(u^4)\\&=(-256\pi^2\log 2+1792\zeta(3))u^3+O(u^4).\end{aligned}\tag{53}$$

The logarithmic terms cancel between (52) and (53), giving

$$S(u)=8-24\pi^2u^2+1792\zeta(3)u^3+O(u^4).$$

This is (3). Since $S^{(r)}(0)=\sum_{k\ge 0}g^{(r)}(k)$, we obtain

$$S'(0)=0,\qquad S''(0)=-48\pi^2,\qquad S'''(0)=6\cdot 1792\zeta(3)=10752\zeta(3),$$

and Theorem 1.1 follows.

Remark 6.1 (Why the third derivative is different). The factor

$$C(u)=\frac{8u^3}{(\frac{1}{2}-u)^3}$$

is forced by the four negative-index Pochhammer factors in (20): three of them contribute $b-a=c-a=d-a=-2u$, while the fourth contributes $e-a=-\varepsilon$. After the reciprocal transformation removes the balancing factor ε, the negative tail begins exactly at order u^3. Thus the constant, first, and second Taylor coefficients are controlled entirely by the bilateral gamma quotient; the cubic coefficient is the first one that sees the reflected tail. This is consistent with the parameter-differentiation and WZ philosophy used by Hou and Sun [2], but the tail calculation above is the additional ingredient needed at third order.

7. Analytic justifications

For completeness, we record the convergence facts used above.

(1) Formula (2) shows that $g(k+u)$ is a rational factor in $k+u$ times $(27/64)^k$ and a quotient of three Pochhammer symbols. Stirling's formula, uniformly on compact u-sets avoiding poles, proves local uniform convergence of S and of every fixed number of its u-derivatives.

(2) In (6), after the specialization (8), the ratio of successive absolute values tends to $27/64$ locally uniformly in (u, ε). The rational weight (9) has polynomial growth in k. This proves the dominated-convergence step leading to (14).

(3) The bilateral identity (16) and the reciprocal identity (23) are first applied with $0 < u < \delta$ and $\varepsilon > 0$, where their convergence hypotheses hold and no denominator vanishes. The function $D = \varepsilon\mathcal{B}$ is analytic at $\varepsilon = 0$ by (30), so the Laurent expansion in Section 4 is legitimate.

(4) The kth summand in (43) is $O(k^{-2+6u}\log k)$ for real u near zero, and its first u-derivative is $O(k^{-2+6u}\log^2 k)$. Choosing, for example, $|u| < 1/12$ gives summable majorants independent of u. Hence H_1 is continuously differentiable at zero, so $H_1(u) = H_1(0) + O(u)$, and (44) follows by termwise passage to the limit.

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